

Required Exercises

Exercise 1

Consider the following context-free grammar G:

$$S \rightarrow SS+ \mid SS* \mid a$$

1. Is the string $a + a * a$ in $L(G)$? [10 points]
2. Give a leftmost derivation for the string $aa * aa + *$. [10 points]
3. Give a rightmost derivation for the string $aa * aa + *$. [10 points]
4. Give a parse tree for the string $aa * aa + *$. [10 points]
5. Give an equivalent grammar without immediate left recursions. [10 points]

1. 不符合文法，该文法只能接受单独的a或者以+或*结束的字符串。

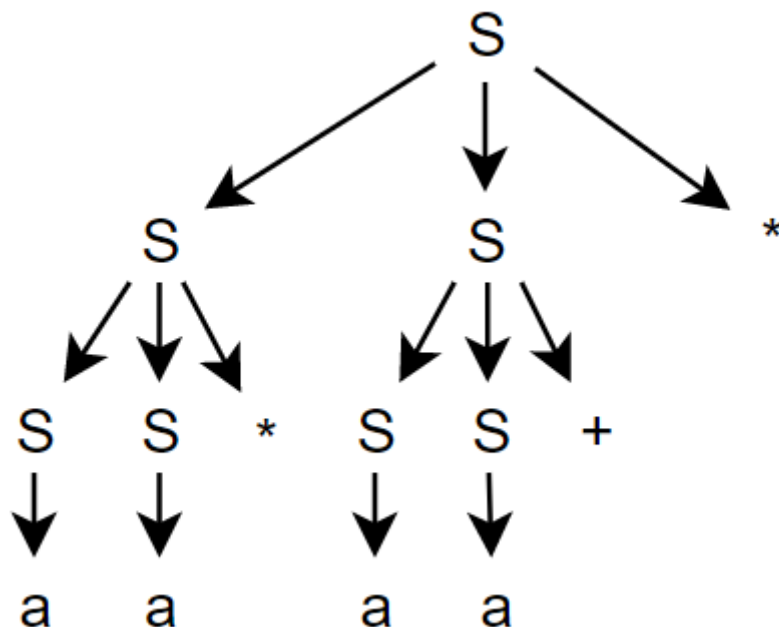
2.

$S \rightarrow SS* \rightarrow SS*SS* \rightarrow aS*SS* \rightarrow aa*SS* \rightarrow aa*SS+* \rightarrow aa*aS+* \rightarrow aa*aa+*$

3.

$S \rightarrow SS* \rightarrow SSS+* \rightarrow SSa+* \rightarrow Saa+* \rightarrow SS*aa+* \rightarrow Sa*aa+* \rightarrow aa*aa+*$

4.



5.

$S \rightarrow aS'$
 $S' \rightarrow S+S' \mid S*S' \mid \epsilon$

Exercise 2

Consider the following grammar G:

$$S \rightarrow aB$$

$$B \rightarrow S + B | \epsilon$$

1. Construct the predictive parsing table for G. Please put down the detailed steps, including the calculation of FIRST and FOLLOW sets. [25 points]
2. Is the grammar LL(1)? [5 points]
3. Can an LL(1) parser accept the input string $aaaa + ++$? If yes, please list the moves made by the parser; otherwise, state the reason. Before parsing, please resolve conflicts in the parsing table if any. [20 points]

1.

FIRST sets

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FIRST(S) = {a}
FIRST(B) = FIRST(S) ∪ {ε} = {a, ε}
FIRST(+) = {+}
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FOLLOW sets

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FOLLOW(S) = FIRST(+) ∪ FOLLOW(B) = {+, $}
FOLLOW(B) = FOLLOW(S) = {+, $}
```

for $S \rightarrow aB$

$FIRST(aB) = \{a\}$, 那么添加 $S \rightarrow aB$ 到 $M[S, a]$

for $B \rightarrow S+B$

$FIRST(S+B) = \{a\}$, 那么添加 $B \rightarrow S+B$ 到 $M[B, a]$

for $B \rightarrow \epsilon$

$FIRST(\epsilon) = \{\epsilon\}$, $FOLLOW(B) = \{+, \$\}$, 那么添加 $B \rightarrow \epsilon$ 到 $M[B, +]$, 添加 $B \rightarrow \epsilon$ 到 $M[B, \$]$,

Non-terminal symbol	Input symbols		
	a	+	\$
S	$S \rightarrow aB$		
B	$B \rightarrow S+B$	$B \rightarrow \epsilon$	$B \rightarrow \epsilon$

2. 是LL(1)文法, 因为预测表中不存在包含两个及以上产生式的单元格
3. 该LL(1)分析器可以接受串 $aaaa + ++$ 。

步骤	分析栈	输入栈	产生式或匹配
1	$\$S$	$aaaa+++ \$$	$S \rightarrow aB$
2	$\$Ba$	$aaaa+++ \$$	"a"匹配

步骤	分析栈	输入栈	产生式或匹配
3	\$B	aaa+++ \$	$B \rightarrow S+B$
4	\$B+S	aaa+++ \$	$S \rightarrow aB$
5	\$B+Ba	aaa+++ \$	"a"匹配
6	\$B+B	aa+++ \$	$B \rightarrow S+B$
7	\$B+B+S	aa+++ \$	$S \rightarrow aB$
8	\$B+B+Ba	aa+++ \$	"a"匹配
9	\$B+B+B	a+++ \$	$B \rightarrow S+B$
10	\$B+B+B+S	a+++ \$	$S \rightarrow aB$
11	\$B+B+B+Ba	a+++ \$	"a"匹配
12	\$B+B+B+B	+++ \$	$B \rightarrow \epsilon$
13	\$B+B+B+	+++ \$	"+"匹配
14	\$B+B+B	++ \$	$B \rightarrow \epsilon$
15	\$B+B+	++ \$	"+"匹配
16	\$B+B	+ \$	$B \rightarrow \epsilon$
17	\$B+	+ \$	"+"匹配
18	\$B	\$	$B \rightarrow \epsilon$
19	\$	\$	"\$"匹配

Optional Exercises

Exercise 1

Consider the following context-free grammar:

短语 $\rightarrow$ 人 | 短语动词短语

动词 $\rightarrow$ 喜欢 | 讨厌

人 $\rightarrow$ 你 | 我 | 他

The grammar can produce sentences such as “我喜欢你”. Is the grammar ambiguous? If yes, please give one sentence and its multiple parse trees. If no, state the reason. [5 points for the yes/no answer and 15 points for the justification]

存在二义性，如下图所示，左图表示“我喜欢（你讨厌他）”；右图表示“我喜欢你并且讨厌他”。

